

1. Administrative & Study Identification

Full Study Title: A Phase 2, Randomized, Double-Blind, Placebo-Controlled Study to Investigate the Efficacy and Safety of PF-07275315 and PF-07264660 in Adult Participants with Moderate-Severe Atopic Dermatitis **Short Title/Acronym:** A Study To Learn About Two Study Medicines (PF-07275315 And PF-07264660) In People Who Have Moderate To Severe Atopic Dermatitis **Protocol Number:** C4531002 **NCT Number:** NCT05995964 **Sponsor Name:** Pfizer **Investigational Product:** PF-07275315 (Tilrekimig) and PF-07264660 **Phase of Development:** Phase 2 **Medical Monitor:** Pfizer R&D Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of PF-07275315 and PF-07264660 compared to placebo in improving the signs and symptoms of moderate to severe Atopic Dermatitis (AD) at Week 16. **Secondary Objectives:** To evaluate the safety, tolerability, and additional clinical efficacy endpoints (such as vIGA score reduction) of the two investigational medicines across different monthly dosing regimens.

2.2 Background & Rationale To address the ongoing need for improved disease control in patients with moderate to severe atopic dermatitis. AD is a chronic inflammatory skin disease driven by Type 2 immune responses. Tilrekimig (PF-07275315) is an investigational trispecific antibody designed to simultaneously inhibit IL-4, IL-13, and TSLP. Similarly, PF-07264660 inhibits IL-4, IL-13, and IL-33. By targeting multiple downstream and upstream pathways involved in Type 2 (Th2) inflammation, the study aims to deliver improved efficacy over standard-of-care therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 340 participants **Number of Sites:** Multicenter / Global **Estimated Study Start:** 22-Aug-2023 **Estimated Primary Completion:** 13-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Clinical diagnosis of chronic atopic dermatitis for at least 6 months prior to Day 1, with diagnosis confirmed by photograph. 3. Moderate to severe AD defined as having an affected Body Surface Area (BSA) $\geq 10\%$, validated Investigator Global Assessment (vIGA) ≥ 3 , and Eczema Area and Severity Index (EASI) ≥ 16 at both screening and baseline visits. 4. Inadequate response to standard of care topical treatments for at least 4 consecutive weeks within 6 to 12 months, or documented reason why topical treatments are medically inappropriate. 5. BMI of 17.5 to 40 kg/m² and a total body weight >45 kg.

4.2 Exclusion Criteria 1. Significant allergic or autoimmune diseases other than AD (e.g., SLE, Type 1 diabetes, IBD, Multiple Sclerosis). 2. Any active or chronic infection requiring systemic antimicrobial therapy within 60 days prior to Day 1, or active skin infection requiring systemic treatment within 2 weeks prior to Day 1. 3. Evidence of active or latent Tuberculosis (TB), HIV, Hepatitis B, or Hepatitis C infection. 4. History of significant allergic reactions, anaphylaxis, or hypersensitivity to protein therapeutics or to the excipients of the formulated drug products.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple dose levels (low, middle, high) evaluated across study stages, administered once-monthly. **Route of Administration:** Subcutaneous injections (in-clinic). **Duration of Treatment:** 16 weeks of treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Standard bland emollients for skin hydration. **Prohibited:** Systemic immunosuppressant treatments, phototherapy (narrowband UVB or broadband), or regular use of tanning booths within 4 weeks prior to Day 1.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, EASI/vIGA/BSA baseline scoring, TB/Viral screening
Baseline	Day 1	Randomization, First Dose, Baseline Efficacy Assessment, Vital Signs
Interim	Weeks 4, 8, 12	Safety Labs, Efficacy Assessment (EASI/vIGA scoring), Study Drug Administration
End of Study	Week 16	Final Efficacy Assessment (EASI-75 evaluation), Safety Review, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants achieving $\geq 75\%$ reduction in the Eczema Area and Severity Index (EASI-75) at Week 16 compared with placebo. **Measurement Method:** Clinical assessment utilizing the standardized EASI scoring tool.

7.2 Secondary & Exploratory Endpoints * Proportion of participants achieving a validated Investigator Global Assessment (vIGA) score of 0 (clear) or 1 (almost clear) with a ≥ 2 -point improvement. * Percent change from baseline in EASI score. * Incidence of Treatment-Emergent Adverse Events (TEAEs) and Serious Adverse Events (SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection through clinical examinations, patient reports, vital sign monitoring, and clinical laboratory testing at all study visits. **Serious Adverse Events (SAEs):** Reported by the investigator within 24 hours of becoming aware of the event. **Data Monitoring Committee (DMC):** External Data Monitoring Committee to ensure ongoing subject safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy endpoints; Safety Analysis Set for safety endpoints. **Sample Size Justification:** $N=340$ appropriately powered to detect statistically significant, placebo-adjusted improvements in the primary endpoint (EASI-75) across multiple dosing groups. **Handling of Missing Data:** Non-Responder Imputation (NRI) for binary endpoints (e.g., EASI-75 response) and Mixed-Effect Model Repeated Measure (MMRM) applied for continuous efficacy endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Centralized, remote, and routine onsite monitoring by the sponsor to ensure protocol compliance and data accuracy. **Audit Trail:** Secure electronic Case Report Forms (eCRF) utilizing automated edit checks, query generation, and comprehensive audit trailing for clinical data management.

11. Study Identifiers & Governance

Primary ID: C4531002 **Registry Number:** NCT05995964 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study To Learn About Two Study Medicines (PF-07275315 And PF-07264660) In People Who Have Moderate To Severe Atopic Dermatitis **Official Regulatory Title:** A Phase 2, Randomized, Double-Blind, Placebo-Controlled Study to Investigate the Efficacy and Safety of PF-07275315 and PF-07264660 in Adult Participants with Moderate-Severe Atopic Dermatitis

12. Clinical Framework (Atopic Dermatitis Specific)

Primary Condition: Moderate to Severe Atopic Dermatitis **Diagnosis Threshold:** BSA $\geq 10\%$, vIGA ≥ 3 , EASI ≥ 16 **Medical Methodology:** Targeted Biologic Therapy (Trispecific Antibodies inhibiting Th2 pathways) **Optimization Target:** Achievement of EASI-75 response at Week 16

13. Study Procedures & Methodology

Management Pathway: Targeted inhibition of downstream (IL-4/IL-13) and upstream (TSLP/IL-33) inflammatory cytokines. **Education Protocol (HCP):** Mandatory training on standardized clinical scoring tools (EASI, vIGA, BSA). **Education Protocol (Patient):** Counseling on compliant emollient application and avoidance of prohibited systemic/topical treatments. **Quality Audit Mechanism:** Centralized photographic review and automated clinical data queries to ensure scoring consistency.

14. Observation & Follow-Up Schedule

Total Duration: 16 Weeks (Treatment) + End of study Follow-up period **Onsite Visit Cadence:** Baseline, Weeks 4, 8, 12, 16 **Remote Monitoring Frequency:** As indicated for safety follow-up between scheduled visits **Checklist Review Interval:** Adverse event and concomitant medication review at each clinic visit (Q4W)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				